Chatbot to chat with medical books using Retrieval-Augmented Generation Model

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***Abstract*—This research paper presents a novel AI tool aimed at revolutionizing medical education by addressing the specific needs of medical students, fostering effective learning, and managing stress. The implementation leverages cutting-edge technology known as Large Language Models (LLMs), with a focus on utilizing ChatGPT4, Streamlit, and Trulens. The digital assistant serves as a friendly and accurate resource, responding promptly to complex medical queries, thereby optimizing study time. The study meticulously examines the balance between technical innovation and practical benefits, emphasizing the project's goal of not only creating a sophisticated tool but also providing tangible assistance to medical students.**

The report outlines the successful development and deployment of the AI-powered study buddy, showcasing its potential to significantly improve learning environments, particularly in demanding fields such as medicine. Notably, the project emphasizes the importance of groundedness in its chatbot, achieving an impressive accuracy rate of approximately 84% for a single textbook. The integration of Streamlit, ChatGPT4, and Trulens contributes to creating a seamless and user-friendly experience for medical students.

This research underscores the broader implications of AI in education, demonstrating how advancements in technology can positively impact learning outcomes. By reducing the time spent on information retrieval, the AI study buddy empowers medical students to concentrate more on core learning objectives, ultimately facilitating their journey to becoming proficient and successful medical practitioners.

***Index Terms*—AI, medical education, medical students, ChatGPT4, Streamlit, Trulens.**

I. Introduction

In response to the pressing challenges faced by medical students, including escalating workloads and a concerning rise in mental health issues, this research endeavors to introduce and analyze the impact of an innovative solution – MedBuddy, a cutting-edge Medical Chatbot powered by AI. The motivation behind the development of MedBuddy stems from a genuine concern for the well-being and learning experiences of medical students, as underscored by alarming statistics revealing a mental health crisis within medical education [1]. Recognizing the transformative potential of Large Language Models (LLMs), such as ChatGPT4, in addressing these challenges, this research explores how AI technology can be harnessed to create a supportive learning environment for medical students.

The primary objective of MedBuddy is to alleviate the stress and workload burden on medical students by providing accurate and timely answers to complex medical questions. Beyond mere efficiency, the chatbot is strategically designed to contribute to the improvement of mental health outcomes among medical students. This research emphasizes the crucial balance between technical innovation and practical benefits, asserting that MedBuddy is not just a sophisticated tool but a tangible support system tailored to the unique needs of medical education.

By delving into the potential of LLMs, particularly in the medical field, this research elucidates the role of these AI superstars in revolutionizing medical education. LLMs, exemplified by GPT-4, have the capacity to transform the way medical knowledge is acquired, disseminated, and applied. The paper examines how these models can enhance education, facilitate medical research, support clinical decision-making, and contribute to improved patient care.

However, the incorporation of AI in medical education is not without challenges, particularly in terms of ensuring trustworthiness. The paper acknowledges the risk of misinform- motion and knowledge gaps and proposes strategic steps to address these concerns, including data quality, model validation, human oversight, explainable AI (XAI), and transparent communication.

Amidst these considerations, MedBuddy emerges as a potential game-changer in medical education, promising to not only lighten the academic load but also bridge the AI-human understanding gap. By providing accessible, reliable, and up-to-date medical knowledge, MedBuddy aims to empower

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medical students from diverse backgrounds, aligning with the ideals of equal opportunity in education.

As the paper progresses, it will delve into the development process, functionalities, and the potential impact of MedBuddy, backed by empirical evidence and insights. The research anticipates that the successful integration of AI into medical education, as exemplified by MedBuddy, could pave the way for transformative advancements in learning environments and contribute to the creation of a more resilient and diverse future healthcare workforce.

II. Literature Survey

The literature survey delves into the world of AI chatbots in healthcare and education, scrutinizing 30 research papers to understand their strengths, weaknesses, and potential impact. The analysis encompasses existing applications, their efficacy, and current trends, with the overarching goal of improving AI tools for maximal influence on medical and public health education.

In the realm of public health education, Francesco Baglivo et al. (2023) examined AI chatbots, particularly in the context of vaccination training [2]. Gunther Eysenbach (2023) investigated the role of ChatGPT in medical education, shedding light on its capabilities and authenticity concerns [2]. Taha Bin Arif et al. (2023) discussed the potential impact of ChatGPT on medical research, emphasizing the crucial need for professional verification [3]. Julia Astrid Moldt et al. (2022) explored the technical aspects of AI in medical education, noting the absence of Large Language Model (LLM) usage [4]. Jeong-Won Han, Junhee Park & Hanna Lee (2022) developed an AI chatbot program aimed at enhancing nursing skills related to electronic fetal monitoring [5].

Further exploring the potential applications of AI-powered chatbots in medical education, Nima Ghorashi et al. (2023) provided insights into their various uses [6]. Additionally, Faiza Alam, Mei Ann Lim, and Ihsan Nazurah Zulkipli (2023) delved into the ethical integration of AI in medical education [7]. Alaa A. Qaffas (2019) focused on improving chatbot semantics using Wit.ai [8], while Hyunsu Lee (2023) explored ChatGPT's potential in biomedical education [9]. Chetan Bulla (2020) addressed the language divide in healthcare through an AI-based medical assistant chatbot [10], and David Kovacek and James C L Chow (2021) evaluated an AI-assisted chatbot for radiation safety education [11]. LuXu et al. (2021) conducted a review of current trends in chatbot technology in medicine, particularly in healthcare and oncology applications [12].

Rehan Ahmed Khan et al. (2023) discussed ChatGPT's application in medical education and clinical management [13], while Feng, Songwei; Shen, Yang MD, PhD (2023) considered ChatGPT's potential roles in shaping the future of medical education [14]. Dr. K S Wagh (2020) developed an automated system for education queries [15].

V. Murali, R. et al. (2018) The ChEMBL database offers curated bioactivity data and compound annotations, accessible through various means like web URLs and APIs. This paper introduces "ChEMBL Bot," a chatbot providing a conversational interface to query ChEMBL compounds. Using natural language processing, it interprets user questions in casual language and retrieves relevant information about compound associations with targets, diseases, mechanisms of action, FDA approval, and patents. ChEMBL Bot aims to enhance accessibility and user engagement with ChEMBL's chemical data. [16]

Geethalekshmy et al. (2023) encompasses the development of a personalized assistive application for the elderly, leveraging ontology modeling based on the ICF framework. The study illustrates the ontology's role in capturing a person's functioning, considering personal traits, environment, and health factors. Figures 1 and 2 depict the ontology's structure and instance assertions, while Figure 3 outlines rule-based inference for assistance actions. The system architecture employs semantic reasoning to deliver tailored assistance, utilizing user profiles, activity logs, and medical histories.
Protege´ is utilized for ontology management, while Apache Jena facilitates Semantic Web application development and RDF data inference.[17]

R. S. Nambiar (2023) survey explores existing methodologies in question-answering systems, particularly in the context of pandemic-related inquiries, focusing on pregnant women and neonates during the COVID-19 outbreak. Various approaches, such as text mining, pre-trained models like BERT, and graph-based methods, have been employed. Notably, the Farm-Haystack framework, integrating natural language processing and backend engineering, emerges as a promising solution. However, fine-tuning on domain-specific data is essential for optimal performance. Existing systems often lack specificity for sensitive populations, underscoring the need for tailored solutions. The survey also highlights challenges in topic modeling and sentence embedding, emphasizing the importance of innovative techniques like LDA and Sentence-BERT integration. Overall, while existing frameworks provide a foundation, customization and refinement are crucial for addressing the nuanced needs of specialized domains, such as healthcare for pregnant women and neonates during a pandemic. The remaining studies continue to explore AI chatbots in medical education and patient care, with a focus on specific domains like oncology, public health, and nursing. The findings collectively underscore the need for advanced AI technologies, professional verification, and rigorous evaluation in developing and applying AI chatbots in the medical field [18].

P. C. Nair (2022) delves into the utilization of deep learning methodologies to enhance the functionality of chatbots in HR settings. Emphasizing the importance of effective communication systems in human resources, the study investigates key components including dataset acquisition, feature engineering, experimental analyses, discussions of findings, and prospective directions for further research. Initially, attention is given to the sourcing of data from open-access platforms like Kaggle, facilitating the creation of a comprehensive dataset encompassing a variety of HR-related inquiries and

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responses. Subsequently, the discussion progresses to feature engineering techniques, highlighting the pivotal role of deep neural network architecture in enabling the chatbot's natural language processing capabilities. Techniques such as tokenization, embedding layers, and activation functions are explored, alongside a comparative analysis of different optimizers and weight initializers to optimize model performance. The experimental phase evaluates the chatbot's performance through metrics such as accuracy, precision, sensitivity, and F1 score, with particular focus on the impact of various optimizers and weight initializers. Notably, the Adam optimizer emerges as the most effective, demonstrating the highest accuracy among alternatives. Following this, the discussion section elaborates on the implications of the findings, suggesting avenues for improvement such as incorporating voice models, considering user emotions, and expanding the dataset for enhanced training. Ultimately, the study concludes by underlining the significant potential of deep learning techniques in augmenting HR chatbot functionality and calls for further research to address existing limitations and explore advanced neural network architectures.[19]

In summary, the comprehensive analysis of these 19 research papers reveals a growing interest in AI chatbots across diverse medical and health fields. While the studies showcase the potential benefits, they also highlight existing limitations, emphasizing the need for advanced technologies and meticulous verification processes. This research underscores the urgency for projects like MedBuddy to integrate advanced AI solutions, enhancing accuracy, reliability, and user experience in both medical education and patient care. Ongoing research in this area aims to address these gaps, paving the way for more effective and intelligent AI tools in healthcare.

III. Methodology

A. RAG Model

B. Integration of Retrieval-Augmented Generation (RAG) Model

Retrieval-augmented generation (RAG) is a technique for enhancing the accuracy and reliability of generative AI models with facts fetched from external sources. In other words, it fills a gap in how LLMs work.

The architecture of RAG model is in the Fig. 1.

MedBuddy achieved a significant advancement through the integration of the Retrieval-Augmented Generation (RAG) model. This cutting-edge technology revolutionized MedBuddy's interaction with medical information, enhancing its user-friendliness and efficiency.

1) Enhancements through RAG and Llama Indexing: The combined use of RAG and Llama indexing brought about notable improvements in MedBuddy's capabilities:

a) Intuitive Interface: Uploading medical textbooks in PDF format became seamless, facilitated by an intuitive interface. This user-friendly design simplified the process of providing MedBuddy with the necessary information for learning and growth.

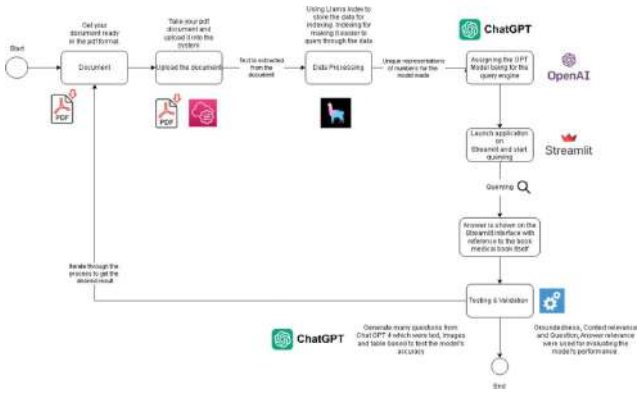

Fig. 1. RAG Architecture

b) Efficient Querying: Empowered by RAG, MedBuddy demonstrated the ability to swiftly and accurately search through uploaded data, delivering pertinent information promptly. Say goodbye to sifting through lengthy documents – MedBuddy brought answers directly to the user.

c) Minimal Preprocessing: The smart system design required minimal preparation before MedBuddy could commence learning from new data. This streamlined approach resulted in faster processing times and expedited access to essential information.

d) Speedy Retrieval: Llama indexing played a pivotal role in accelerating data retrieval. This powerful technology ensured MedBuddy could swiftly and efficiently access requested information, saving valuable time.

Imagine having a vast library of medical knowledge at your fingertips, easily accessible with a few clicks. This is the transformative impact of RAG and Llama indexing in action. These technologies have positioned MedBuddy as an efficient tool for medical learning and support, providing instant access to essential information.

By seamlessly integrating these advanced technologies, MedBuddy has evolved into a more powerful and user-friendly resource for those seeking accurate and up-to-date medical knowledge. With improved speed, efficiency, and a user-friendly interface, MedBuddy is poised to revolutionize the way we learn and access medical information.

2) User Interface: A simple and intuitive interface (Fig. 2) that allows medical students to interact with the chatbot, submit queries, and receive responses using Streamlit.

3) Backend Services: Includes server-side components for query processing (Fig. 3), data handling, and maintaining the chatbot's responsiveness and efficiency.

C. User Interaction

MedBuddy's user interaction is streamlined to ensure ease of use:

1) Query Submission: Users can submit queries related to medical topics, which are processed by the chatbot.

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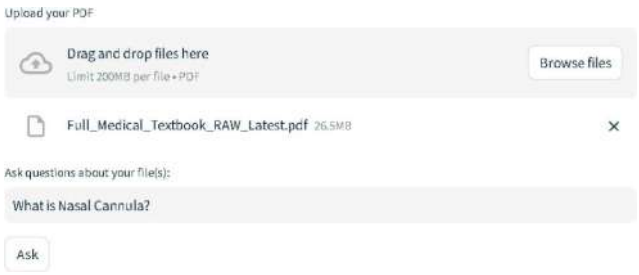

Fig. 2. Uploading File & Asking Question

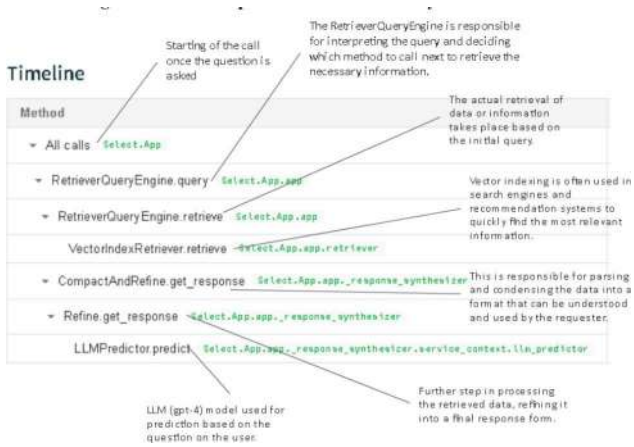

Fig. 3. RAG Behind the scene

2) Response Generation: The chatbot, powered by the RAG model, generates responses that are accurate and contextually relevant.

IV. Results

1) RAG Model Test: In the test the model was successfully being able to extract information from the pdf provided as showin in Fig. 4.

Fig. 4. RAG model tested

2) Performance Metrics and Evaluation: Performance metrics for MedBuddy include:

a) Groundedness: Measures the correctness of the information provided in response to queries. The groundedness scores are shown in Fig. 5.

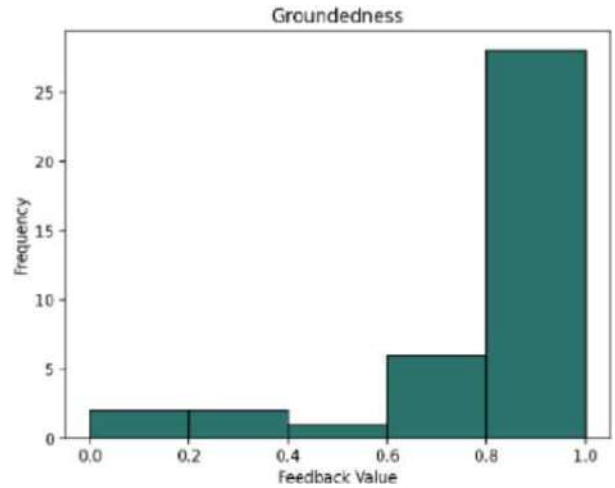

Fig. 5. RAG Groundedness

b) Response Time: Evaluates the speed at which the chatbot provides responses.

c) Context & Answer Relevance: Determines how well the chatbot's responses match the context of the queries. Regular evaluations based on these metrics ensure the chatbot's effectiveness and guide further enhancements. The contet relevance is shown in Fig. 6 and answer relevance Fig. 7.

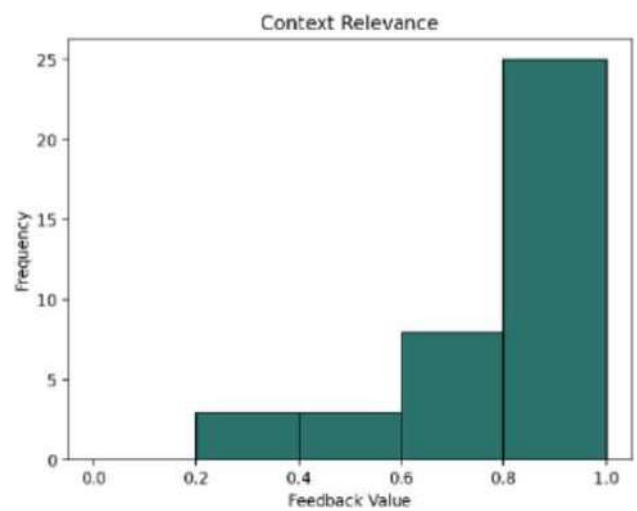

Fig. 6. RAG Context Relevance

V. Conclusions and Future Work

The development of the MedBuddy chatbot marks a landmark moment in the integration of Artificial Intelligence (AI),

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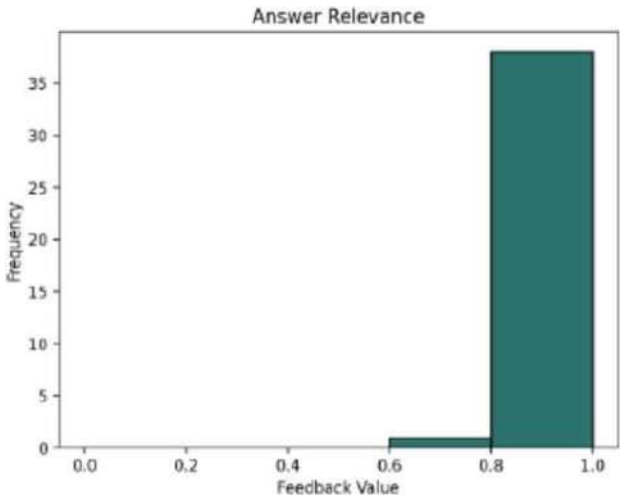

Fig. 7. RAG Answer Relevance

particularly Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG), in medical education. This project set out to create a tool that not only revolutionizes the learning experience for medical students but also contributes to the broader landscape of technological innovation in healthcare education.

The project's journey was one of continuous learning and adaptation. Initial attempts at fine-tuning a pre-trained LLM with medical textbooks yielded valuable insights, paving the way for the transition to the more effective RAG model. This journey highlighted the intricacies and immense potential of AI applications in medical education. The culmination of these efforts - MedBuddy - stands as a testament to the feasibility and effectiveness of using advanced AI models to deliver accurate, reliable, and contextually relevant medical information.

MedBuddy's impact transcends mere academic assistance. It embodies an equal opportunity tool that fosters the aspirations of future medical professionals. By alleviating the academic burden with its readily accessible knowledge base, MedBuddy plays a crucial role in mitigating the stress often associated with medical education. This, in turn, empowers students to focus their energy on deeper understanding and personal growth.

Furthermore, MedBuddy democratizes access to medical knowledge, particularly for those facing geographical or resource constraints. By breaking down barriers and providing an accessible, AI-powered learning companion, MedBuddy paves the way for a more inclusive and equitable future in medical education.

In conclusion, MedBuddy stands as a beacon of innovation in the field of medical education. Its development and implementation demonstrate the immense potential of AI technology to revolutionize learning experiences and empower the next generation of healthcare professionals. By fostering a more accessible and inclusive educational landscape, MedBuddy paves the way for a brighter future in healthcare and beyond.

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